

Type Modifiers

A variable type modifier allows you to adjust a variable type, eg

```
short int x; // a 2 byte integer (range from -32768 to 32767)
```

Here "short" is a type modifier that reduces the size of "int"

The following type modifiers will be discussed in this lecture:
short, long, unsigned, const -- the typical ranges are given in

https://www.tutorialspoint.com/cplusplus/cpp_data_types.htm

Other type modifiers will be discussed later in the term

short: makes the data type smaller (less bytes, smaller range)
if possible -- useful to reduce memory consumption, eg

```
short int x; // 2 byte integer on 32/64 bit CPU
```

long makes the data type larger (more bytes, larger range)
if possible -- useful to increase range of variable type

```
long int y; // doubles the byte size of int on some CPU types
```

a long / short type modifier will change the size of a variable if
possible -- in some cases the compiler will ignore the request, eg

```
sizeof(long int) == sizeof(int) for a 32/64 bit compiler
```

unsigned: restricts the range of a variable to positive or zero numbers. this also typically doubles the positive range, eg

unsigned int z; // range from 0 to 4294967295 on 32/64 bit CPU
(size in bytes remains unchanged with unsigned)

unsigned char w = 237; // range, 0 to 255 (char is -128 to 127)



the main application of unsigned is to double the range and avoid negative numbers in applications where they are not valid

const: fixes a variable to be a constant (ie can't be changed from its declared initialization value), eg

const double PI = 3.14159;

PI = 1.7; // this can't be done -- generates a compiler error

More than one modifier can be applied to a variable type, eg.

```
const unsigned int q = 12345; // constant positive int
```

```
unsigned long int p = 1999; // unsigned large int
```

```
const unsigned short int s = 270; // constant unsigned short int
```


